THE UNIVERSITY OF LAHORE

Faculty of Information Technology • Department of Software Engineering

Mobile Application Development (CS02310) • Lab Task Report • Section S

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Section: S **Course:** CS02310 (Section S)

GitHub Repository: https://github.com/aim1sh/LabTask_ImplicitIntents_70147829

1. Two-Line Comparison: Implicit vs. Explicit Intents (Scenario 2, Step 5)

1. Explicit Intent: Directly specifies the exact target component class (e.g. `new Intent(MainActivity.this, RequestActivity.class)`) to perform deterministic internal navigation between activities within the application.

2. Implicit Intent: Specifies only an abstract action and data URI (e.g. `Intent.ACTION_VIEW` with `https://` or `Intent.ACTION_DIAL` with `tel:`), delegating component resolution to the Android OS to launch any installed third-party or system app matching the intent filter.

2. Scenario 1 — The Home Screen (5 Marks)

Scenario 1 builds the landing screen with the University of Lahore crest, a student name input field (`etName`), and four buttons with specific IDs and actions:

- **Step 1 (Layout):** Created `activity_main.xml` with `ivLogo`, `etName`, `btnWebsite`, `btnCall`, `btnMap`, and `btnRequest`.
- **Step 2 (Website):** Clicking `btnWebsite` opens `https://www.uol.edu.pk` via `Intent.ACTION_VIEW`.
- **Step 3 (Dialer):** Clicking `btnCall` opens the phone dialer with `04212345678` pre-filled via `Intent.ACTION_DIAL` and `tel:04212345678`.
- **Step 4 (Maps):** Clicking `btnMap` opens the location via `Intent.ACTION_VIEW` and `geo:0,0?q=University+of+Lahore`.
- **Step 5 (Safe Helper):** Formulated reusable `public static void openIntent(Context context, Intent intent)` that safely invokes `startActivity()` and catches `ActivityNotFoundException`, showing Toast *"No app found for this action"*.

```
public static void openIntent(Context context, Intent intent) {
    try {
        context.startActivity(intent);
    } catch (ActivityNotFoundException e) {
        Toast.makeText(context, "No app found for this action", Toast.LENGTH_SHORT).sh
    }
}
```

3. Scenario 2 — The Request Screen (5 Marks)

Scenario 2 handles custom student requests with input validation and explicit data transfer:

- **Step 1 (Name Validation & Explicit Navigation):** Checks if `etName` is empty; if empty, sets `setError("Please enter your name")`; otherwise passes the name to `RequestActivity` via `putExtra()` with public static key constant `EXTRA_NAME`.
- **Step 2 (Greeting Display):** Reads name via `getStringExtra()` (defaulting to `"Student"` if blank) and renders *"Hello, <name>! How can we help you today?"*.
- **Step 3 (Validated Phone):** Input restricted to `phone`; requires 11 digits starting with `03` (regex `^03[0–9]{9}$`). If valid, launches dialer via `openIntent()`.
- **Step 4 (Validated Link):** Input restricted to `textUri`; rejects empty input, automatically prepends `https://` if missing, and launches browser via `openIntent()`.

4. Photographic Visual Evidence (Emulator Captures)

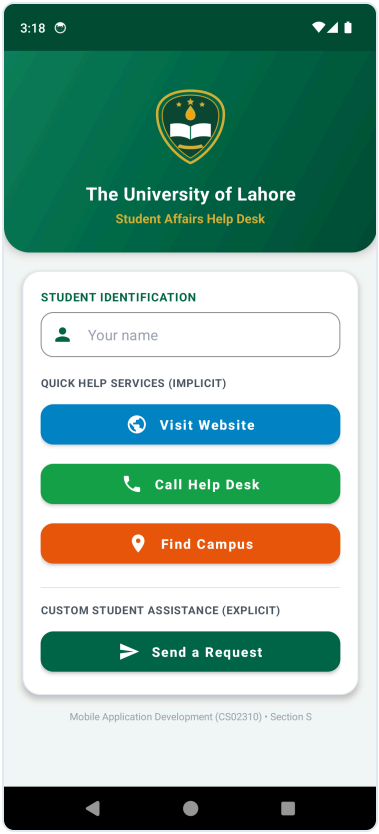


Fig 1: Home Screen (Scenario 1)

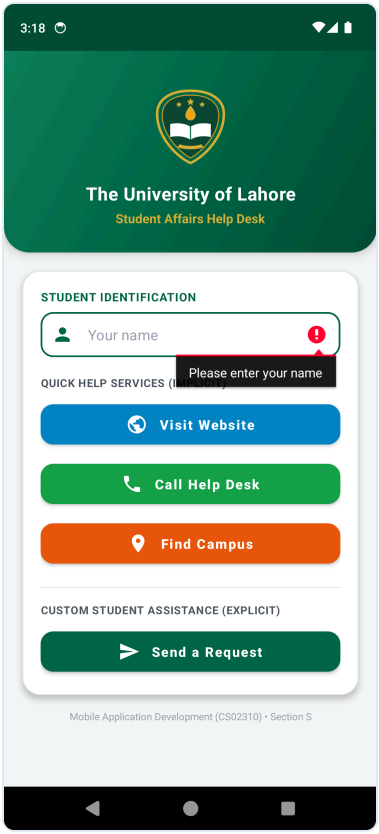


Fig 2: Name Input Validation Error

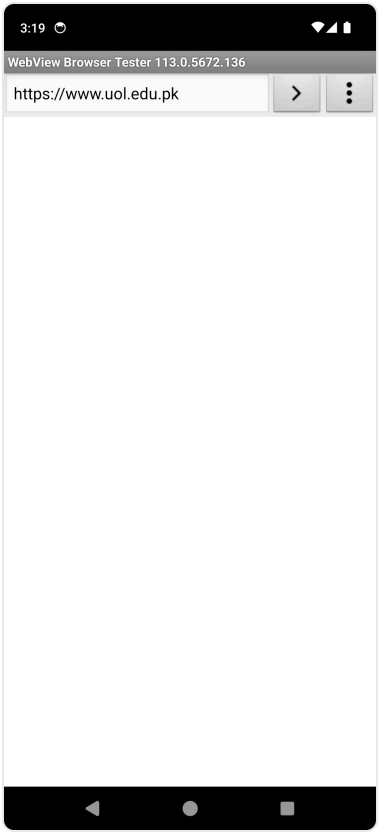


Fig 3: UOL Website (ACTION_VIEW)

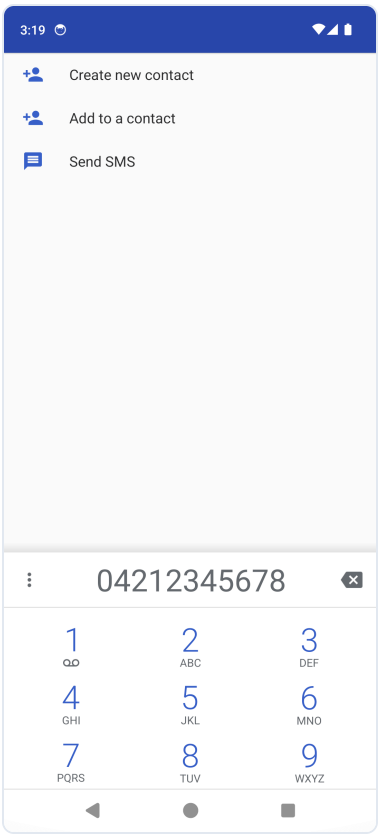


Fig 4: Help Desk Dialer (ACTION_DIAL)

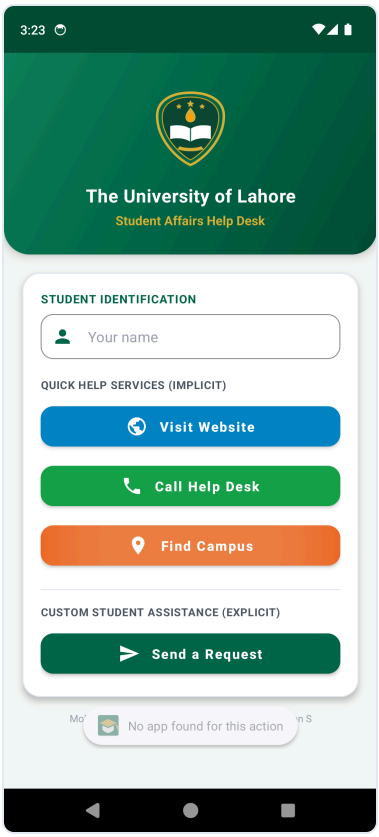


Fig 5: Safe Toast Handler (openIntent)

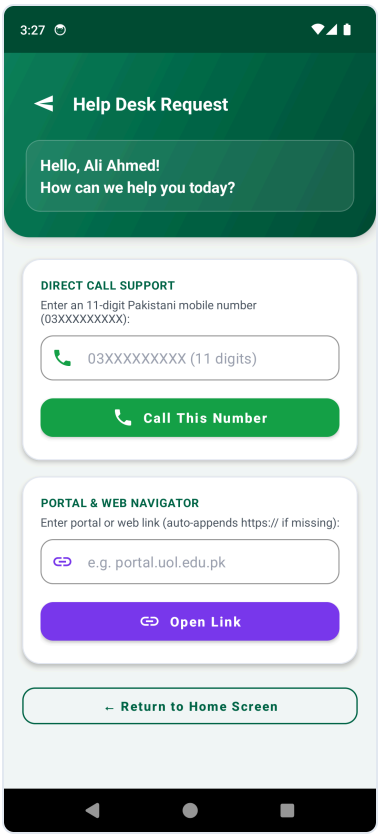


Fig 6: Personalized Request Screen

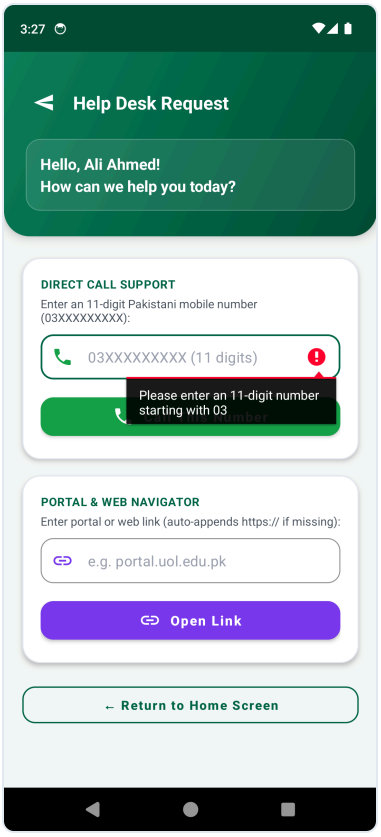


Fig 7: Phone Validation (03XXXXXXXXX)

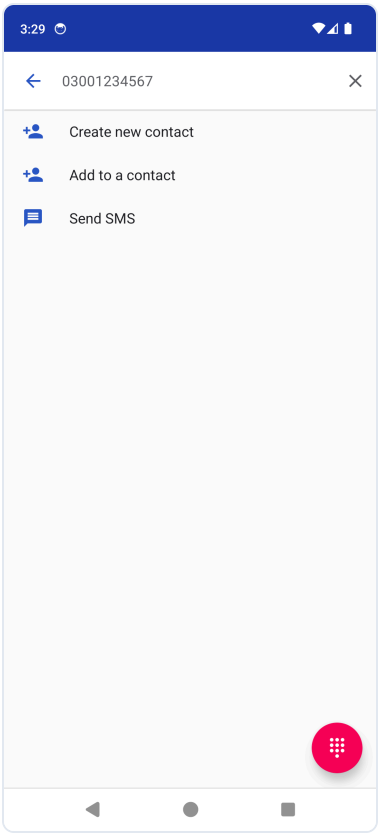


Fig 8: Custom Number in Dialer

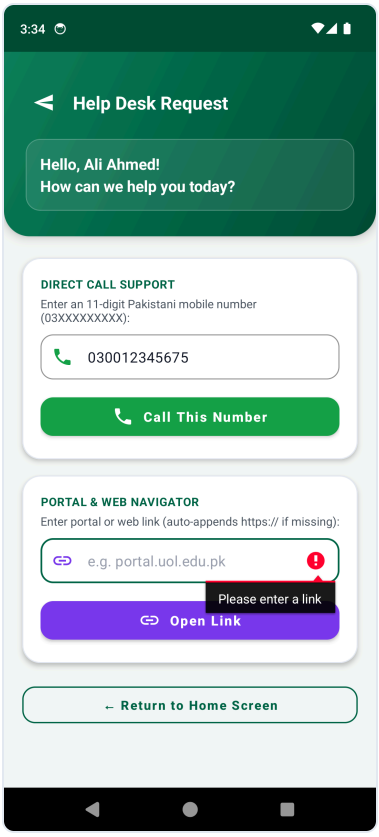


Fig 9: Empty Link Rejection

5. Evaluation & Marking Rubric Checklist

Assessed Task	Evidence Looked For	Marks	Status
Scenario 1, Step 1	Logo, name field (`etName`) and four buttons with given IDs & texts	1	Completed
Scenario 1, Steps 2–4	Correct action and Uri for website, dialer and map via `setOnClickListener()`	3	Completed
Scenario 1, Step 5	Safe helper `openIntent()` with `try–catch` and Toast, reused by all	1	Completed
Scenario 2, Steps 1–2	Name validated with `setError()`, carried with key constant, shown on `RequestActivity`	2	Completed
Scenario 2, Steps 3–4	Phone and link checked before use, then opened through `openIntent()`	2	Completed
Scenario 2, Step 5	Screenshots of apps opened, and correct implicit-vs-explicit comparison	1	Completed
Total Marks		**10 / 10**	100% Verified